

Cybersecurity Awareness and Digital Safety Practices Among Adolescents Aged 16–20 in Dhangadhi, Nepal



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ABSTRACT

The rapid growth of internet access and smartphone usage has transformed how adolescents communicate, learn, and interact. Social media, messaging apps, and online platforms are now integral to daily life, offering educational and social opportunities but also exposing users to cybersecurity risks such as phishing, malware, scams, identity theft, and privacy breaches.

Adolescents aged 16–20 are among the most active online users, making them particularly vulnerable to digital threats. In Nepal, internet penetration and social media use among youth have increased rapidly; however, many adolescents still lack sufficient cybersecurity awareness and safe online practices.

This study aims to assess cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal. It will examine adolescents' understanding of common cybersecurity threats, evaluate their online safety behaviors, and analyze how factors such as digital literacy and internet usage influence awareness. A descriptive research design will be applied, collecting data from at least 384 participants through structured questionnaires distributed across schools, colleges, community settings, and online platforms.

The findings are expected to provide insights into the current level of cybersecurity awareness, identify gaps between knowledge and actual online behavior, and offer recommendations for educators, policymakers, and community organizations to strengthen cybersecurity education and promote responsible digital practices.

Keywords: Cybersecurity Awareness, Digital Literacy, Online Safety, Cybercrime

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Chapter 1

INTRODUCTION

1.1 Background of the study

Nepal is experiencing a rapid digital transformation as internet access and smartphone usage continue to expand across the country. Digital technologies are increasingly integrated into everyday activities such as communication, education, online banking, and access to information. Social media platforms, messaging applications, and online learning systems have become essential tools, especially for young people. While these technologies provide numerous benefits, they also expose users to various cybersecurity risks such as phishing attacks, online scams, malware, identity theft, and privacy breaches.

Adolescents aged 16–20 represent one of the most active groups of internet users in Nepal. Many students regularly use digital platforms for social networking, entertainment, academic activities, and communication with peers. However, despite their frequent use of technology, many adolescents lack adequate awareness of cybersecurity threats and safe online practices. For example, some young users may reuse weak passwords, click on suspicious links, share personal information online, or use unverified applications without understanding the potential risks involved. These behaviors increase their vulnerability to cyber threats and online fraud.

In recent years, Nepal has witnessed a significant increase in internet penetration and the adoption of digital services. Platforms such as mobile banking applications, social media networks, and online learning systems are widely used among students. At the same time, reports of cyber-related incidents such as phishing scams, account hacking, and online fraud are also increasing. These trends highlight the importance of improving cybersecurity awareness among internet users, particularly adolescents who are highly active in digital environments.

Despite the growing use of digital technologies, cybersecurity education and awareness programs remain limited in many schools and communities in Nepal. Many adolescents use the internet daily but may not fully understand how to protect their personal information, recognize cyber threats, or adopt safe digital behaviors. Without proper awareness and guidance, they may unknowingly expose themselves to cybersecurity risks that can lead to financial loss, data breaches, or emotional distress.

Dhangadhi, as a rapidly developing urban center in western Nepal, reflects these broader digital trends. Adolescents in this region increasingly rely on smartphones and internet services for academic activities, communication, and entertainment. However, there is limited research examining the level of cybersecurity awareness and digital safety practices among adolescents in this area. Understanding their knowledge, experiences, and online behaviors is essential for identifying gaps in awareness and promoting safer digital practices.

Therefore, this study aims to assess cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal. By examining adolescents' understanding of cybersecurity threats and their online safety behaviors, the research seeks to identify knowledge gaps and recommend practical strategies to improve cybersecurity awareness and promote responsible digital behavior among young internet users.

1.2 Literature Review

(Bhattacharya et al., 2025) refers that, Cybersecurity safeguards systems, networks, and programs to prevent digital attacks that seek to gain unauthorized access, manipulate, or compromise sensitive data. (Wongmahesak et al., 2025) enhances digital literacy and reduce cybercrime fear, emphasizing customized educational approaches for various age groups, including community workshops, school integration, and online resources.

(Nayyar & Gupta) highlights the growing importance of cybersecurity in education as digital tools and virtual learning environments become more widespread. It emphasizes the need for cyber awareness, secure practices, and integrating cybersecurity education into the curriculum to protect students and institutions from emerging threats. Overall, it stresses building a culture of safety to ensure a secure digital future for learning. (Hussain & Phulpoto, 2024) examines how digital literacy empowers individuals across different groups through surveys, interviews, and literature analysis. Findings show that higher digital literacy improves job opportunities, social participation, and overall digital engagement while highlighting challenges like access, skills gaps, and the digital divide. The research also proposes training programs and policy support to strengthen digital literacy in communities.

(Prakash & Sanju, 2024) evaluates how effectively social media can be used to promote cyber safety awareness by analyzing public knowledge, attitudes, and online behaviors. It also examines demographic factors to determine how social media campaigns can be better designed for different groups.

(Bayzan et al.) examines youth-focused digital literacy programs and assessment tools in Canada to address growing concerns about online safety, privacy, and cyber risks. An environmental scan identified 15 existing programs, while a literature review found 12 assessment tools. Based on these findings, a new adaptable digital literacy program and a 15-item assessment tool were developed to strengthen youth skills in digital fluency, safety, ethics, and consumer awareness. The research highlights the need for targeted digital literacy education to better prepare youth for the digital world.

Cybercrime has emerged as a major threat in the digital era, affecting individuals, organizations, and governments worldwide. As digital dependence increases, users with limited cybersecurity awareness and digital literacy become more vulnerable to cyber threats. Understanding the nature of cybercrime is therefore essential for promoting safe digital practices. (Kaspersky, 2025) defines cybercrime as criminal activities that target or use computers, networks, or digital devices, primarily for financial gain, data theft, or system disruption. The study categorizes cybercrime into various forms, including phishing, identity theft, malware attacks, ransomware, cyberextortion, and distributed denial-of-service (DDoS) attacks. It highlights that cybercriminals often exploit users' lack of awareness by using deceptive techniques such as fraudulent emails, fake websites, and social engineering tactics. The report further emphasizes that cybercrime is increasingly affecting not only large organizations but also individuals and small businesses, leading to financial loss, reputational damage, and loss of trust in digital systems. Additionally, it stresses that the rise in cybercrime is closely linked to insufficient cybersecurity awareness and poor digital practices among users. The study recommends adopting preventive measures such as using strong passwords, avoiding suspicious links, keeping software updated, and protecting personal information online. These recommendations align with the objectives of the present study, as they highlight the role of cybersecurity awareness and digital literacy in reducing cyber risks and encouraging safe digital behavior. The findings support the need for educational and awareness-based interventions to empower users to protect themselves from cybercrime in an increasingly digital society.

Children represent one of the most vulnerable groups in the digital world. According to (Innocenti, 2021a), children increasingly use the internet for playing, learning, and socializing. However, this exposure brings risks such as cyberbullying, online predators, and misinformation. UNICEF emphasizes that digital literacy from an early age is essential to balance opportunities with risks. Similarly, the UNICEF (Innocenti, 2021b) found that while children benefit from online learning and communication, weak cybersecurity measures expose them to privacy breaches, exploitation, and harassment.

Teenagers face distinct challenges as they navigate social media and online communities. Information leaks often result in harassment and bullying, with long-term psychological impacts. A study by (Sun et al., 2025) examined how cyber moral literacy influences cyberbullying behaviors among late adolescents. The research explored three critical questions:

- a) How does cyber moral literacy influence cyberbullying behaviors?
- b) To what extent does moral disengagement mediate the relationship between cyber moral literacy and cyberbullying?
- c) How does guilt moderate the relationship between cyber moral literacy and cyberbullying?

Findings suggest that teenagers with stronger moral literacy are less likely to engage in cyberbullying, while moral disengagement increases harmful behaviors. Guilt acts as a moderating factor, reducing the likelihood of harassment when adolescents recognize the ethical implications of their actions. This underscores the importance of integrating moral and ethical dimensions into cybersecurity education for teenagers.

Adults, though generally more aware of the consequences of unsafe practices, often ignore them due to convenience or affordability issues. The use of pirated products, whether entertainment or productivity software, poses serious risks. It is found that pirated software often contains hidden malware, spyware, or backdoors that compromise personal and organizational data, exposing users to ransomware attacks and system instability. Similarly, the (Partida, 2020) reported that cracked or counterfeit software bypasses security protocols, making systems more vulnerable to cyberattacks. Despite awareness, many adults continue to rely on pirated products, highlighting the gap between knowledge and practice.

The study “Perceived Digital Threats Influencing Smartphone Use Among the Aging Population” by R. Kanimozhi, V. Padmavathi, and P. S. Ramesh highlights a critical issue: many older adults are not sufficiently prepared to use modern technology with confidence. As the world rapidly digitizes, the global rise in the elderly population has been accompanied by increasing adoption of smartphones. These devices are not just tools of convenience, they help seniors maintain relationships, access healthcare, and engage in recreational activities, ultimately improving their overall quality of life. However, the research reveals a concerning trend. A significant number of elderly individuals are stepping into the digital world without fully understanding the risks and challenges of cybersecurity. While smartphones open doors to connection and independence, they also expose older adults to threats such as privacy breaches, fraud, and data misuse.(Kanimozhi et al., 2025)

Cybersecurity awareness plays a crucial role in reducing cybercrime and promoting safe digital practices, particularly among general users. In developing countries like Nepal, the lack of awareness regarding online threats significantly increases users’ vulnerability to cyber fraud, identity theft, and online scams. The (NepalPolice, 2021) highlights the importance of adopting basic cyber safety practices to prevent cybercrime. The guidelines emphasize avoiding suspicious links and emails, not sharing personal information or private photos online, and being cautious of fraudulent job offers and online scams. The Bureau also stresses the use of strong passwords, regular password changes, and enabling multi-factor authentication (MFA) to protect online accounts. Additionally, users are advised to avoid conducting sensitive financial transactions on public Wi-Fi networks and to regularly update devices with reliable antivirus software. Furthermore, the Cyber Bureau underscores the importance of data protection through regular backups and encourages families to educate children and household members about online risks. These preventive strategies demonstrate that cybercrime prevention is not solely dependent on technical security systems but also on user awareness and responsible digital behavior. This source is particularly relevant to the present study as it provides a local and practical perspective on cybersecurity awareness, supporting the need for educational initiatives to promote safe online practices and reduce cybercrime in Nepal.

1.3 Research Gap Analysis

Existing literature has highlighted the increasing challenges of cybercrime and the important role of cybersecurity awareness and digital literacy in promoting safe online behavior. Several studies have examined cybersecurity issues among young internet users and emphasized the importance of digital literacy, ethical awareness, and responsible online practices. For example, Sun et al. (2025) examined the relationship between cyber moral literacy and cyberbullying behavior among adolescents, while Wongmahesak et al. (2025) highlighted the role of educational programs in improving digital literacy and cybersecurity awareness. Similarly, Bhattacharya et al. (2025) emphasized the importance of protecting digital systems and data from cyber threats in an increasingly connected world.

Despite these contributions, several important research gaps remain.

1.3.1 Limited research in the context of Nepal

Most existing studies on cybersecurity awareness have been conducted in technologically advanced or global contexts. There is relatively little empirical research focused on developing countries such as Nepal, where levels of digital literacy, technological access, and cybersecurity awareness vary widely. Although organizations such as Nepal Police Cyber Bureau (2021) have issued general guidelines on cyber safety, academic studies that examine the level of cybersecurity awareness and digital safety practices among young internet users in Nepal remain limited. This lack of localized research creates a gap in understanding how adolescents in Nepal perceive and respond to cyber threats.

1.3.2 Lack of focused research on adolescents' cybersecurity practices

While previous studies have examined digital literacy and cyber risks among different populations, limited research has specifically focused on adolescents' everyday digital behavior and cybersecurity practices. Adolescents aged 16–20 represent one of the most active groups of internet users, particularly on social media platforms, messaging applications, and online learning systems. However, there is insufficient research examining how adolescents understand cybersecurity threats such as phishing, online scams, identity theft, and privacy risks, as well as how they apply safe digital practices in their daily online activities.

1.3.3 Insufficient evidence on the relationship between cybersecurity awareness and online behavior

Many studies discuss the importance of cybersecurity awareness, but fewer studies analyze the relationship between adolescents' awareness and their actual online behavior. In many cases, individuals may have basic knowledge of cyber risks but still engage in unsafe practices such as reusing weak passwords, sharing personal information online, or clicking on suspicious links. Therefore, there is a need for research that examines not only the level of awareness but also how that awareness influences adolescents' digital safety practices.

1.3.4 Lack of localized studies in semi-urban regions

Most cybersecurity awareness research has been conducted in large urban centers or developed countries. There is limited research examining cybersecurity awareness among adolescents in semi-urban areas such as Dhangadhi, Nepal, where access to digital technologies is increasing but structured cybersecurity education may still be limited. Understanding the local context is important because digital behavior may be influenced by factors such as access to technology, education level, and social environment.

Given these research gaps, the present study aims to examine cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal. By analyzing adolescents' knowledge of cybersecurity threats and their online safety behaviors, the study seeks to provide context-specific insights that can support educators, policymakers, and community organizations in developing effective cybersecurity awareness programs for young internet users.

1.4 Objectives

1.4.1 General Objective

To assess cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal.

1.4.2 Specific Objectives

To measure the level of cybersecurity awareness among adolescents aged 16–20.

To identify common cybersecurity threats encountered by adolescents on digital platforms.

To examine the relationship between digital literacy and cybersecurity awareness among adolescents.

To analyze the digital safety practices adopted by adolescents to prevent or respond to cybersecurity threats.

To provide recommendations for improving cybersecurity education and promoting safe digital behavior among adolescents.

1.5 Hypothesis

H1: Adolescents with higher digital literacy demonstrate greater awareness of cybersecurity threats compared to adolescents with lower digital literacy.

H2: Adolescents who have greater awareness of cybersecurity threats are more likely to adopt safe digital practices.

H3: Adolescents who spend more time online or use social media frequently are more likely to be exposed to cybersecurity risks.

1.6 Conceptual Framework

The conceptual framework of this study illustrates the relationships between adolescents' demographic factors, digital exposure, cybersecurity awareness, and digital safety practices. It assumes that several independent variables—such as age, gender, education level, time spent online, social media usage, digital literacy, and previous experience with online threats—can influence adolescents' understanding of cybersecurity.

Cybersecurity awareness acts as a mediating variable connecting these independent factors to the outcome variable, digital safety practices. Adolescents with higher cybersecurity awareness are more likely to adopt safe online behaviors, including protecting personal information, avoiding suspicious links, reporting scams, and using secure platforms.

By analyzing these relationships, the study aims to identify which factors most strongly predict cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal. This framework provides the basis for understanding how

personal and digital exposure factors interact with awareness to shape safe online behavior.

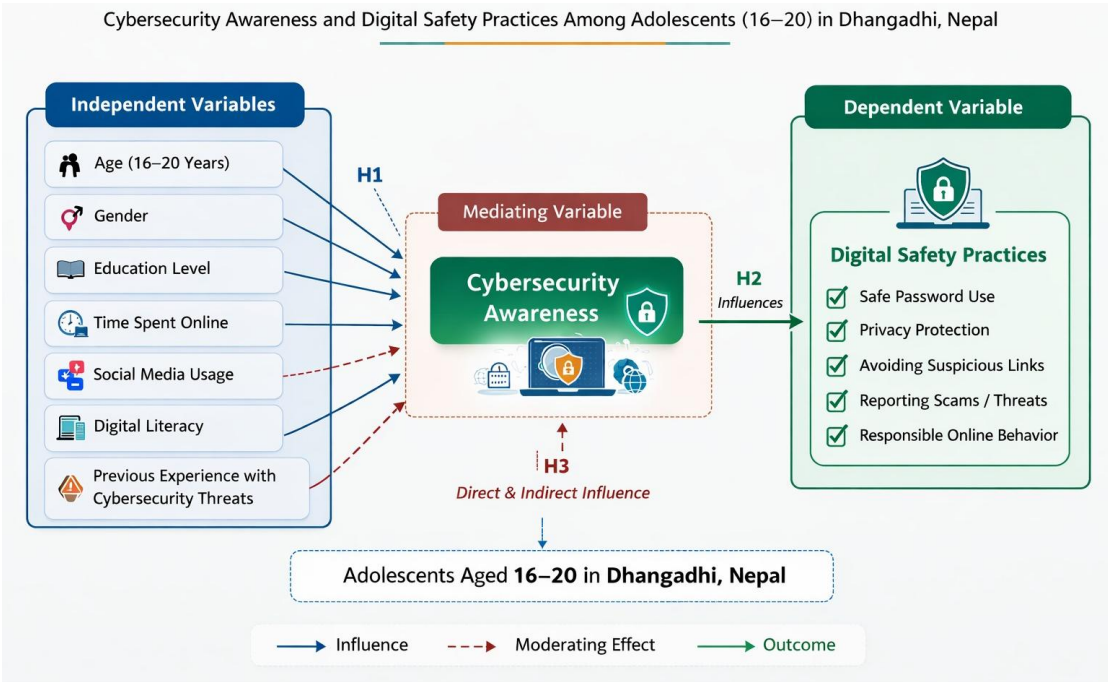


Figure 1.6 Conceptual Framework

Chapter 2

METHODOLOGY

2.1 Description of the Research Station

This study will be conducted with the support of the National Academy of Science and Technology, which provides essential resources for research on cybersecurity awareness. The institution offers access to computers, reliable internet connectivity, and a comfortable workspace for preparing questionnaires, managing responses, and analyzing data. Continuous guidance from supervisors will ensure that the study maintains academic standards and methodological rigor throughout the research process.

2.2 Research Design, Sample Size, and Sampling Strategies

A descriptive research design will be employed to investigate cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi.

Sample Size: The study will include 384 participants, providing a representative understanding of adolescents' digital behaviors and awareness levels.

Data Collection Methods: Structured questionnaires will be administered through:

- Schools and colleges (offline)
- Community meetups (offline)
- Online platforms (e.g., Google Forms)

Sampling Strategies:

Convenience Sampling: Engaging participants who are readily accessible during visits.

Purposive Sampling: Selecting adolescents within the 16–20 age range to ensure the relevance of responses.

This mixed approach ensures inclusion of adolescents with diverse educational backgrounds and online experiences, resulting in a comprehensive dataset.

2.3 Study Area

The study will be conducted in Dhangadhi, targeting locations where adolescents aged 16–20 can be reached. These include:

- Selected schools and colleges

- Community centers and local meetups

Online social media platforms to include adolescents who may not be reachable offline

By combining offline and online approaches, the study captures a wide range of adolescent experiences, ensuring a comprehensive assessment of cybersecurity awareness and digital safety practices in the area.

2.4 Research Tools, Techniques, and Data Collection Strategies

Data for this study will be collected using a structured questionnaire, specifically designed to assess cyberbullying awareness and digital safety practices among adolescents aged 16–20.

Online Distribution: The questionnaire will be shared via Google Forms, enabling a wider reach and convenience for participants.

Offline Collection: Additional data will be gathered through community visits, school interactions, and college meetups. This approach allows the researcher to engage directly with adolescents who may not have online access.

By combining online and offline methods, the study ensures a more representative and balanced dataset. The questionnaire will be designed to be simple, clear, and easy to understand, facilitating accurate and honest responses from participants.

2.5 Data Management and Analysis Strategies

After data collection, responses from both online forms and field visits will be carefully organized to maintain accuracy and confidentiality. The following steps will be undertaken:

Data Cleaning: Removing incomplete or inconsistent responses.

Data Organization: Categorizing and coding responses for systematic analysis.

Secure Storage: Storing all data in password-protected files to ensure participant privacy.

Analysis Techniques:

Descriptive Statistics: Summarizing numerical data such as awareness levels and frequency of digital safety practices.

Thematic / Pattern-Based Analysis: Examining open-ended responses to identify common experiences and insights regarding cyberbullying and online safety.

The analysis will focus on interpreting results clearly and objectively, reflecting the real experiences of adolescents and highlighting key patterns in cybersecurity awareness and digital safety behavior.

Chapter 3

EXPECTED OUTCOME AND IMPACT OF THE RESEARCH

This study is expected to provide a comprehensive understanding of the current level of cyberbullying awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal. By collecting responses from participants through structured questionnaires, school and college visits, community interactions, and online forms, the research will offer an accurate picture of how adolescents recognize cyberbullying, behave online, and adopt preventive measures.

The outcomes will highlight not only the level of awareness but also the gaps between knowledge and actual practices, such as sharing passwords, ignoring privacy settings, responding inappropriately to online harassment, or engaging in risky digital behavior.

Major Expected Outcomes

a) Assessment of Cyberbullying Awareness Among Adolescents

The study will evaluate how well adolescents identify and understand cyberbullying. Findings will indicate whether higher digital literacy correlates with better awareness and safer online practices.

b) Analysis of Digital Safety Practices

The research will document common preventive behaviors, including using strong passwords, reporting abusive content, avoiding risky online interactions, and managing privacy settings. It will also identify unsafe practices and potential risk factors that increase adolescents' vulnerability to cyberbullying.

c) Identification of Factors Influencing Awareness and Safety

The study will examine how variables such as digital literacy, social media usage, previous cyberbullying experience, gender, and education level influence awareness and online safety behaviors. This will help identify adolescents who may require targeted guidance or interventions.

d) Evidence-Based Recommendations for Awareness Programs

Practical strategies will be suggested to improve cyberbullying awareness and digital safety among adolescents.

Recommendations may include:

- Integrating short digital safety lessons into school curricula.
- Organizing peer-led awareness sessions in colleges.
- Conducting community workshops for parents and youth.
- Sharing simple educational content via social media.

e) Contribution to Safer Digital Environments

The research will provide actionable insights for educators, policymakers, and community organizations, enabling them to design interventions that promote responsible and safe online behavior among adolescents in Dhangadhi.

Chapter 4

APPLICATION OF THE PROPOSED RESEARCH

The findings of this research will have practical applications that can directly benefit the local community, educational institutions, and individuals of all age groups. Since the study focuses on understanding cybersecurity awareness across children, teenagers, adults, and older people, the insights gained can guide meaningful improvements in how digital safety is taught and practiced in everyday life.

One of the main applications of this research will be in educational institutions, such as schools and colleges. Since a portion of the data will be collected from these settings, the results will help teachers and administrators recognize the digital vulnerabilities present among students. The research can support the creation of short cybersecurity modules that can be integrated into computer classes, extracurricular sessions, or orientation programs. Simple lessons on password management, recognizing scams, safe browsing habits, and basic privacy protection can be implemented with minimal additional resources. This makes the research highly relevant for institutions that want to improve student safety without needing major investments.

Another important application lies within community-level programs. Because part of the study involves community visits and in-person interactions, the findings can inspire local organizations, youth groups to organize small awareness sessions. These could be conducted in community halls, ward offices, or even informal gatherings. The research will highlight which groups require more support and what specific digital threats they are most vulnerable to. This will help community leaders design targeted sessions instead of generic trainings that may not address the actual needs of the population.

The research will also be useful in understanding how economic factors influence unsafe digital behavior, particularly the use of pirated software, unverified apps, or free websites that contain hidden risks. By documenting these issues, the study can support the development of practical alternatives, such as promoting free and legitimate digital tools, encouraging the use of secure platforms, or teaching people how to verify whether a website or download is genuine. This makes the findings valuable for policymakers, government bodies, and digital literacy campaigns.

In addition, the study has strong applications in public awareness initiatives. Once the patterns and common risky behaviors are clearly identified, the information can be

simplified into posters, flyers, or social media content to reach a wider audience. Local stakeholders can use the results to create short messages like “Do not share passwords,” “Avoid suspicious links,” or “Use trusted apps only,” making cybersecurity education visible and easy to understand. These messages can be spread through community pages, public events, or academic networks.

Chapter 5

DELIVERABLES OF THE RESEARCH PROJECT

The research project will produce the following key deliverables:

Deliverables	Descriptions
Research Report / Thesis	A complete written report containing all chapters, findings, analysis, and recommendations.
Research Article	A brief article summarizing the major results and insights for sharing with academic or community platforms.
Presentation Slides	A concise slide deck highlighting the research process, data, and conclusions for academic or community presentations.
Awareness Materials	Simple posters, messages, or guidelines that can be used by schools and communities to promote safe digital practices.
Survey Dataset and Questionnaire	An anonymized dataset from Google Forms and field surveys, along with the final questionnaire for future reference.

Table 5.1 Deliverable of the research

These deliverables ensure that the research does not stay theoretical but can be reused, presented, and applied in multiple practical settings.

Chapter 6

PROJECT PLANNING AND MILESTONES

This chapter presents the overall project plan, including the duration of the study, tentative starting and ending dates, and key milestones. The total duration of the proposed research is 4 months, starting from January 2026 and ending in April 2026.

The research activities are divided into several phases based on the objectives and methodology of the study. The table below presents the Gantt chart of the overall research project, highlighting monthly activities and major milestones.

Research Activities	Jan 01- 15	Jan 16- 30	Feb 01- 15	Feb 16- 30	Mar 01- 15	Mar 16- 30	Apr 01-15	Apr 16-30
Proposal Submission								
Literature Review								
Questionnaire and Methodology Design								
Survey and Data collection								
Data organization and analysis								
Report Writing								
Review and Feedback								
Presentation								
Final report submission								

Table 6.1 Gantt Chart

This project plan provides a clear roadmap for completing the research within four months. The defined milestones and monthly schedule will help monitor progress and ensure that all stages, from proposal finalization to report submission, are completed efficiently. By following this plan, the research will be conducted systematically and yield reliable results.

Chapter 7

BUDGET DISTRIBUTION PLAN

This chapter presents an estimated budget distribution plan covering all major research-related activities. The budget has been prepared with careful consideration of the scope of the study, available resources, and the need to minimize costs while ensuring effective data collection, analysis, and dissemination.

7.1 Budget Estimation

The estimated costs include expenses related to proposal preparation, fieldwork, data collection, data analysis, documentation, dissemination activities, and essential research materials. All amounts presented below are approximate and based on current local prices in Nepal.

S.N.	Budget	Description	Estimated Cost
1.	Field work and community visits	Local travel, communication, and coordination expenses	15,000
2.	Research Materials	Textbooks, policy reports, research journals, and access to online academic databases	8,000
3.	Equipment and Accessories	Essential research hardware such as storage devices, and computer devices (Depreciation 20%) required during data collection and virtual interview	4,000
4.	Awareness and seminar dissemination	Posters, banners, small seminar or presentation materials	9,500
5.	Printing & Stationery	Printing of reports, questionnaires, consent forms, and stationery	10,000

6.	Software and Tools	For accessing of survey platforms, and cybersecurity assessment tools required for data analysis and interpretation, according to requirement	5,000
7.	Contingency	Miscellaneous and unexpected costs	5000
Total			56,500

Table 7.1 Cost Estimation

7.2 Funding Sources

The primary source of funding for this research is expected to be the Research and HRD Department of NAST.

Chapter 8

RESEARCH DISSEMINATION PLAN

Dissemination of research findings is an important part of this study, as it ensures that the knowledge gained is shared beyond the written report and contributes to academic discussion and public awareness.

a) Oral Presentation

The key findings of this research will be shared through oral presentations at community, schools, and colleges. These presentations will focus on the research objectives, methodology, major findings, and practical recommendations. Oral presentation allows to interact with peoples and researcher which enhances the effectiveness of the finding of the research.

b) Journal Publication

Based on the finding of the study, a research article will be prepared and submitted to a Research and HRD Department of National Academy of Science and Technology. The article will emphasize the local context of Dhangadhi city and present evidence-based recommendations that may be useful for similar communities facing digital literacy challenges. Journal publication will ensure that the research contributes to the academic literature and reaches a global audience.

c) Poster and Leaflet Presentation

Poster presentations will be prepared for academic conferences, seminars, and institutional research events. The posters will visually present the background of the study, research objectives, methodology, key findings, and suggested awareness strategies in a clear and simple format.

This method is particularly useful for communicating research outcomes to a broad audience, including students, educators, and non-technical participants. It is possible that to produce simple infographics in local languages (Tharu/Doteli) to distribute in the community.

d) Digital Media

The article helps to spread the awareness through social media like Facebook, TikTok to share short clips demonstrating how to identify and be aware about digital scams.

Through oral and poster presentations, journal publication, and newspaper articles, this research aims to reach both academic and non-academic audiences. The dissemination plan ensures that the research findings are not limited to a written report but are actively shared to promote cybersecurity awareness, improve digital literacy, and encourage safe digital practices within the community and beyond.

Chapter 9

RELEVANCE TO THE UN SUSTAINABLE DEVELOPMENT GOALS

The proposed research on Cybersecurity Awareness and Digital Literacy Strategies for Promoting Safe Digital Practices directly contributes to the achievement of the United Nations Sustainable Development Goals (SDGs). By focusing on improving digital literacy and cybersecurity awareness among different age groups in Nepal, the study supports inclusive, safe, and sustainable digital development.

a) SDG 4: Quality Education

SDG 4 aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. This research strongly aligns with this goal by addressing digital literacy as an essential component of modern education. In Nepal, especially in regions like Dhangadhi, many citizens lack basic knowledge about safe digital practices despite increasing access to the internet and digital services. Children, teenagers, adults, and elderly individuals often use digital platforms without understanding online risks such as phishing, identity theft, cyberbullying, or data misuse.

b) SDG 8: Decent Work and Economic Growth

Cybersecurity awareness protect digital assets and mobile banking (e.g., e-Sewa, Khalti), which are essential for small-scale entrepreneurship and financial inclusion in Nepal.

c) SDG 9: Industry, Innovation, and Infrastructure

Target 9.c. focuses on universal and affordable access to the internet. This research ensures that "access" is meaningful by providing the necessary safety knowledge, as technical access without literacy leads to increased vulnerability to cybercrime.

d) SDG 16: Peace, Justice, and Strong Institutions

Target 16.1 & 16.2: By investigating strategies to prevent online scams, cyberbullying, in the digital realm.

The proposed research supports sustainable development by empowering individuals with the knowledge needed to safely navigate the digital environment. By improving digital literacy and cybersecurity awareness at the community level, the study helps

reduce digital inequality, promotes responsible technology use, and strengthens trust in digital systems. These contributions align with the broader vision of the UN SDGs to create inclusive, resilient, and sustainable societies.

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